

3D Objects: Compare and Classify

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Name: _____

Date: _____

Learning Intention

I can compare and classify 3D objects by their features and explain why particular shapes are suited to different real-world uses.

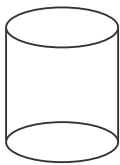
Strategy Box: Look, Touch and Test

Faces are flat surfaces. Curved surfaces can roll. Edges are where faces meet.

Vertices are corner points. Compare how objects stack, slide and roll.

Classify using a clear rule, then explain why every object fits.

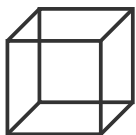
Worked Example - Why Is a Can a Cylinder?



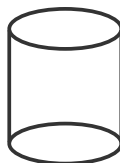
A cylinder has 2 flat circular faces and 1 curved surface.

It can stand, stack and roll, so it suits a drink can.

A. Warm Up - Identify the Features



Cube



Cylinder



Cone

1. Faces: ____ Edges: ____ Vertices: ____ 2. Flat faces: ____ Curved surfaces: ____ 3. Flat faces: ____ Vertices: ____

Picture Practice - Sort by Movement

P1. Write cube, cylinder or cone in every box where it belongs.

Can roll

Can stack

Can slide

Quick check: One object may belong to more than one group.

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B. Build Your Skills - Compare Objects

4. How are a cube and rectangular prism alike? _____

5. How are a cylinder and cone different? _____

6. Which has no flat face: sphere / cone / cube? Explain. _____

7. Which objects can both roll and stand? _____

C. Classify with Your Own Rule

8. Objects: basketball, cereal box, party hat, soup can, dice.

My sorting rule: _____

Group 1: _____ Group 2: _____

Why the rule works: _____

D. Apply It - Shape Suited to Use

9. Why is a cube useful for building blocks?

10. Why is a sphere useful for a ball?

11. Design a container. Choose a 3D shape and explain why its features suit the use.

Learning check

Accept accurate feature language and classifications supported by a clear rule.

A. Warm Up and Picture Practice

1. Cube: 6 faces, 12 edges, 8 vertices.
2. Cylinder: 2 flat faces, 1 curved surface.
3. Cone: 1 flat face, 1 vertex.

P1. Roll: cylinder, cone. Stack: cube, cylinder. Slide: all three.

B. Compare Objects - Sample Answers

4. Both have flat faces, edges and vertices; both stack and slide.
5. A cylinder has 2 flat faces and no vertex; a cone has 1 flat face and 1 vertex.
6. Sphere; its whole surface is curved.
7. Cylinders can roll on the curved surface and stand on a flat face.

C-D. Classify and Apply

8. Answers vary; the rule must place every object consistently.
9. Flat square faces allow cubes to stack without rolling.
10. A sphere rolls smoothly in every direction.
11. Designs vary; check the chosen features match the stated use.

Teacher feedback checklist

- ☐ Names features accurately
- ☐ Compares objects
- ☐ Uses a valid sorting rule
- ☐ Links shape to function
- ☐ Explains reasoning with evidence